Team question bank and architecture card

Read this together once. In the review, answer in **one plain sentence first**, then add detail only if the reviewer asks. If a question belongs to another segment, let that presenter answer it.

The system in one picture

```
Python simulator ──> versioned JSON example bundle ──> browser UI
Lab sender/capture scripts ──> receiver CSV import ─────> browser UI
                                                        │
                                                        ├─ gap and window measurements
                                                        ├─ fixed and learned detector rules
                                                        ├─ charts, tables, reports, and exports
                                                        └─ bit decoder

Browser Web Crypto ──> encrypted bytes ──> timing model ──> receiver unlock attempt
                                        └─> guardian timing check
```

The interface is React/TypeScript. The guided scenes reuse the analysis functions behind the detailed workbench. Data is fetched as a versioned JSON bundle and validated in the browser. CSV parsing runs in a browser worker. The browser uses Web Crypto for AES-256-GCM. Python also contains the simulator, detector reference implementation, tests, and the closed-lab send/capture scripts. The current guided demo does not send network packets.

Core questions

Reviewer asks	Short answer	Best speaker
What is ICMP?	The protocol used by ping; we examine the arrival times of its Echo Requests.	You
What is a covert timing channel?	A sender encodes information in *when* otherwise ordinary packets arrive, rather than in their visible contents.	You
How does zero or one get sent?	A short planned gap means 0, a long planned gap means 1; the receiver compares each received gap with a dividing line.	You / Aarush
Are you changing packet payloads?	The timing-channel idea changes gaps; packet-field equivalence still needs real lab captures and field checks to verify.	You
Why receiver-side timestamps?	The receiver observes the timing after network delay and loss have acted on it.	You
What is jitter?	Variation in network delay from packet to packet; it can blur the short/long difference.	You
What does “ms” mean?	Millisecond: one-thousandth of a second. The subtle demo gaps are 950 and 1050 ms.	You
Why compare a normal reference?	Without ordinary traffic under comparable conditions, an unusual-looking gap has no useful context.	You / Harsh
What is a window?	One group of consecutive requests judged together; the guided case uses 32 requests, giving up to 31 gaps.	Harsh
What is standard deviation?	A measure of how much the gaps wobble around their average.	Harsh

Reviewer asks	Short answer	Best speaker
How is the learned baseline made?	From separate ordinary runs, it learns a 95th-percentile limit for the chosen feature; the guided case tests standard deviation.	Harsh
What is the fixed rule?	In the guided case it alarms if standard deviation exceeds 100 ms or mean gap drifts over 100 ms from one second.	Harsh
Why did the fixed rule get 0% false alarms?	On this default condition it raised no alarms at all, so it also missed all ten message windows.	Harsh
Why can the clean-only baseline be bad?	A threshold learned in a clean network may treat ordinary noisy traffic as suspicious; the default example flags all ten ordinary windows.	Harsh
What does "100% detected" mean here?	Ten of ten message windows in this controlled simulated setting. It is not a universal accuracy claim.	Harsh
What is a false positive?	A normal window wrongly marked suspicious; it costs attention or causes unnecessary intervention.	Harsh
What do ROC and AUC mean?	ROC shows detection versus false alarms while the threshold changes; AUC summarizes that curve. These plots still use simulated data.	Harsh
What is the Stress matrix for?	It checks many combinations of channel subtlety, network variation, loss, detector choice, and metric so we can see failure cases.	Harsh
What does pacing do?	A model holds requests toward regular one-second release times. It may blur hidden timing, but adds delay for ordinary traffic.	Harsh
Is pacing deployed?	No. It is a causal browser model; a real relay needs implementation and measured network tests.	Harsh
What is the adaptive decoder?	It finds a dividing line between observed short and long gaps by clustering them into two groups; the fixed decoder uses one second.	Aarush
What happens if a request is lost?	Sequence numbers let the analysis omit gaps crossing a missing request. Missing bits still hurt recovery. The locked-note model itself adds jitter, not packet loss.	Aarush
Is the encryption real?	Yes. The browser uses Web Crypto AES-256-GCM, with a key derived from the phrase using PBKDF2 and a fresh random salt and IV. The timing delivery is modelled.	Aarush
Is the prefilled phrase secure?	It is a demo phrase. A real system would need a strong secret and a reviewed key-management design.	Aarush
Why does a two-letter note become 46 bytes?	Two text bytes plus 16 salt bytes, 12 IV bytes, and a 16-byte authentication tag.	Aarush
Why around six minutes?	46 bytes make 368 bits; this model uses about one gap per bit at roughly one second per gap.	Aarush
Does encryption hide the timing channel?	It hides note contents. The guardian can still observe abnormal pauses, and noise can still corrupt the encrypted bytes.	Aarush
Why does one wrong bit break decryption?	AES-GCM verifies the whole encrypted package; if the key or authenticated bytes are wrong, it rejects the message.	Aarush
Is there a backend/database?	The demo is a browser app loading a local JSON bundle. Python prepares data and offers lab tools; interactive analysis and encryption run in the browser.	Aarush

Reviewer asks	Short answer	Best speaker
What is the unique contribution?	One inspectable teaching/evaluation workflow joins encoding, receiver evidence, detection errors, defence cost, encrypted content, and provenance. We use established algorithms rather than claim a new cipher.	Aarush
What is your strongest honest result?	The software works on its controlled simulated examples and shows concrete trade-offs. Real-network performance remains unmeasured in the shipped bundle.	Aarush
What is next?	Collect repeated receiver-side ordinary and message runs on a closed lab testbed under clean/noisy/lossy conditions, separate training from testing, then report detection, false alarms, bit recovery, and added delay.	Aarush

Three answers the whole team must say the same way

1. **"Are these real lab measurements?"** "No. The shipped bundle has 610 simulated runs and zero included real receiver captures. The site labels them throughout."
2. **"Did you invent the cipher or detection statistics?"** "No. The originality is in integrating them into a transparent sender–detective–guardian evaluation and teaching tool, and exposing the trade-offs."
3. **"Is this production-ready network security?"** "No. It is a tested teaching and analysis tool. A real deployment needs capture data, field checks, traffic-cost measurements, and security review."

If the panel asks about tests or evidence quality

The local V2 passed 72 UI unit tests, 62 Python tests (2 skipped), a production build and size check, and 32 desktop/mobile browser checks (4 intentionally skipped). Browser checks include navigation, encryption success/failure, accessible controls, and layout widths. These checks support **software correctness**, not real-network accuracy. The evidence page includes report/data exports and clearly distinguishes software behaviour, simulation, and required real measurements.

Terms to avoid mixing up

- **Detection rate** = message windows flagged / all message windows.
- **False-positive rate** = ordinary windows wrongly flagged / all ordinary windows.
- **Bit recovery** = correct bits / expected bits; a high percentage can still leave the whole encrypted note unreadable.
- **Encryption** protects content; **encoding** maps bits into timing; **detection** judges observed timing; **pacing** delays release to reduce that timing pattern.
- **Simulated run** is generated data. **Real capture** requires actual receiver-side packet timestamps and a trustworthy record of how they were collected.